Working paper | Not peer reviewed | 13 October 2026

Four micropeptide-source RNAs fail as a colon cancer recurrence panel, and the marker that survives clears a selection-aware null only narrowly

A panel picked for mechanism rather than by screening, tested against a thousand random panels put through the same machinery — on eighteen recurrences. The one marker that beat it now has a null of its own, and passes by four per cent.

Leo Lee[1]

1 Shanghai High School International Division, Shanghai, China | *Working paper*

PRISM-WP-003 (2026)

137 patients, 18 events **2,000** random comparators **8** figures PRISM-WP-003

ABOUT THIS WORKING PAPER

This paper has not been peer reviewed. PRISM's conflict policy states that no student reviews a manuscript from their own school. The proposed referee is at the author's school and is also a co-author of the author on PRISM-1-004, so he is twice barred from refereeing it. Rather than publish a review the journal's own rules forbid, the paper is posted as a working paper. It can be promoted to a numbered article once an eligible referee files a report, which is the route PRISM-WP-002 took to becoming PRISM-1-004.

The framing of this version is not the author's, and one analysis is not his either. The submitted manuscript carried an AI-assisted abstract, discussion and conclusion and a declaration that the author would rewrite them before submission; that rewrite has not happened. The title, abstract and the prose of sections 5 to 13 were written by Claude (Anthropic) for the journal. None of the author's results were altered and no model of his was refitted.

> The submitted study identified a missing comparison — a random single-gene null for LINC00675 — as the gap that left its headline number uninterpretable. The journal computed it (§4.1, §10, Fig. 8), after requiring the re-implementation to reproduce all four of the author's published single-marker C-indices and all 100 of his per-fold values exactly. That computation is the journal's, not the author's; its code is deposited separately so it can be checked or discarded on its own.

Abstract

Transcripts once classified as long noncoding RNAs can encode functional peptides, and several of those peptides have documented roles in colorectal cancer. This study asked whether the source RNAs of four experimentally characterised micropeptides — LINC00467 (ASAP), ELFN1-AS1 (SMIMP), LINC00675/TMEM238L (FORCP) and HOXB-AS3 — form a useful recurrence panel in colon adenocarcinoma.

They do not. In 137 stage II/III TCGA-COAD patients with 18 disease-free interval (DFI) events, the four-transcript elastic-net Cox panel reached a held-out C-index of 0.609 (95% CI, 0.509–0.717) under five-by-three nested cross-validation repeated over five seeds, below the 0.65 development target set in advance. Against 1,000 random four-gene panels put through the identical procedure, it ranked at the 93.4th percentile, one-sided empirical $p = 0.067$. The prespecified comparison fails at its own threshold.

The individual transcripts are the more useful result. Three of the four discriminated at or below chance for primary DFI: 0.462 for LINC00467, 0.367 for ELFN1-AS1 and 0.515 for HOXB-AS3. Across the three cohorts and four transcripts, 7 of 12 estimates fall below 0.5. Only LINC00675 stood out, at 0.700 (95% CI, 0.583–0.819), and it was the sole transcript retained in the final penalised refit.

The study as submitted had no way to judge that 0.700, because it built a null for the panel and none for a single gene. We computed the missing null: 1,000 genes drawn at random and put through the same folds, the same measurement gate and the same unpenalised Cox model. Restricted to the 279 draws that pass the gate in all 25 folds, as all four candidates do, LINC00675 sits at the 98.9th percentile ($p = 0.014$), just above the null's 99th percentile of 0.699. Correcting for its having been the best of four candidates, a result this good arises by chance **4.2%** of the time. The marker therefore clears its first real test, and clears it narrowly.

Two features of the comparator distributions bear on both tests. Of the 1,000 random panels, 264 (26.4%) returned a C-index of exactly 0.500 because the penalty shrank every coefficient to zero, leaving constant risk; among the 736 that fitted anything, 9.0% matched or beat the candidate panel, a one-sided p of 0.091 rather than 0.067. The same effect is far stronger for single genes, where 710 of 1,000 draws produce no model at all, which is why the unmatched single-gene null ($p = 0.004$) overstates the case and the expression-matched one is the figure to quote.

What remains true is that the panel strategy failed, that most of these mechanistically chosen transcripts carry nothing, and that the one that does still rests on 18 events, shows nothing for overall survival, and has never been tested outside this cohort or against clinical risk factors.

Keywords: colorectal adenocarcinoma; LINC00675; TMEM238L; FORCP; micropeptide-source RNA; disease-free interval; TCGA; negative result.

1. Micropeptide-source RNAs as candidate markers

Small open reading frames in transcripts historically classified as long noncoding RNAs can encode functional peptides. Ribosome-profiling atlases have expanded the catalogue of human translated regions, and proteomic studies have supplied direct evidence for subsets of noncanonical translation products [1,2,3]. These approaches distinguish transcription, translation, peptide detection and biological function. Expression of a source RNA is therefore a tractable candidate biomarker, but it cannot by itself establish the abundance or the activity of the encoded peptide. Everything below concerns RNA.

Several experimentally characterised peptides motivate a colorectal cancer panel. LINC00467-encoded ASAP interacts with ATP synthase and promotes mitochondrial respiration and tumour growth [4]. ELFN1-AS1-encoded SMIMP, identified by functional interrogation of noncanonical open reading frames, interacts with SMC1A and supports repression of cell-cycle inhibitors [5]. LINC00675-encoded FORCP localises to the endoplasmic reticulum, suppresses proliferation and tumorigenicity, and supports apoptosis under endoplasmic-reticulum stress in well-differentiated colorectal models [6]. The HOXB-AS3 peptide antagonises hnRNP A1-dependent pyruvate kinase splicing and metabolic reprogramming [7]. LINC00266-1-encoded RBRP enhances IGF2BP1 recognition of m⁶A-modified transcripts [8].

Mechanistic plausibility is not prognostic performance. Established colon-cancer expression classifications and commercial recurrence assays were both developed against independent cohorts and assessed alongside conventional clinical factors [9,10], a standard no single-cohort analysis can meet. This study tested whether a panel chosen on mechanism rather than on a survival screen would discriminate recurrence in TCGA colon adenocarcinoma, and then asked which individual transcripts carried whatever discrimination appeared.

2. What was fixed in advance, and what was not

The distinction matters more here than the point estimates, so it is stated before the results.

Fixed before the outcome data were examined: the four candidate transcripts and their Ensembl identifiers; the primary endpoint (DFI) and primary cohort (stage II/III); the 0.65 C-index development target; the nested cross-validation design, its grids and its seeds; the 1,000-panel random comparator and its one-sided empirical probability; and the fold-support rules for time-dependent AUC.

Decided after results were seen, and therefore exploratory: the focus on LINC00675; every single-marker contrast; the paired panel-minus-marker difference; and the expression median split. These are labelled post hoc wherever they appear and are not corrected for multiplicity. Readers should treat the prespecified panel test as the result of this study and everything concerning LINC00675 as hypothesis generation.

Added after submission, by the journal rather than the author: the random single-gene null of §10. It was computed because the submitted manuscript identified its absence as the decisive gap, and it is reported with an explicit correction for the post hoc selection it is calibrating. It does not convert the exploratory finding into a prespecified one.

3. Data and cohorts

Bulk expression came from the UCSC Xena TCGA-COAD STAR-Counts matrix [11]. That matrix contains $\log_2(\text{count}+1)$ values, as its dataset metadata specifies; the filename must not be read as indicating untransformed counts. Counts were reconstructed as 2^x-1, retaining fractional values arising from upstream averaging across portions and

aliquots. Clinical outcomes and staging came from Supplemental Table S1 of the TCGA Clinical Data Resource, restricted to COAD [12]. DFI and overall survival used the supplied event indicators and time fields with no outcome relabelling. Times were held in days; displayed months use 365.25/12 days.

Four source RNAs were matched to Ensembl identifiers after version-suffix removal. LINC00266-1/RBRP was considered mechanistically but excluded before any model was fitted, because no stable Ensembl identifier could be resolved in this annotation; the exclusion was logged as missing annotation, never as zero expression. Gene-level abundance does not resolve peptide-encoding isoforms.

Source RNA	Ensembl gene ID	Peptide	Documented biology (from the original studies, not from tl
LINC00467	ENSG00000153363	ASAP	ATP synthase activation, mitochondrial respiration, tumour g
ELFN1-AS1	ENSG00000236081	SMIMP	SMC1A interaction, repression of CDKN1A/CDKN2B [5]
LINC00675 / TMEM238L	ENSG00000263429	FORCP	ER localisation, stress-associated apoptosis, suppressed pr
HOXB-AS3	ENSG00000233101	HOXB-AS3 peptide	Antagonism of hnRNP A1-mediated PKM splicing [7]
LINC00266-1	Not resolved; excluded	RBRP	IGF2BP1-mediated m⁶A recognition [8]

Expression columns were restricted to primary-tumour specimens. The first 12 characters of each barcode defined the patient identifier. Where several primary specimens existed, the lexicographically first barcode was kept, giving one specimen per patient without consulting outcomes. The expression matrix held 514 specimens, comprising 471 primary tumours from 458 unique patients, of whom 456 matched a COAD clinical record. Stage II/III eligibility identified 305 patients; 168 lacked DFI status, leaving 137 patients with 18 DFI events. Reverse Kaplan–Meier median follow-up was 31.05 months.

Cohort	*n*	Events	Follow-up (mo)	Panel C-index	Random-panel percentile	One-sided
Primary DFI, stage II/III	137	18	31.05	0.609 (0.509–0.717)	93.4	0.067
Expanded DFI,	181	21	30.98	0.593 (0.478–0.703)	86.7	0.134

Cohort	*n*	Events	Follow-up (mo)	Panel C-index	Random-panel percentile	One-sided
stages I–IV eligible						
Secondary OS, stages I–IV	428	93	27.01	0.548 (0.487–0.606)	89.9	0.102

Although expanded DFI eligibility admitted stage IV, no stage IV patient had usable DFI, so that cohort is stages I–III. Differences between endpoints cannot be separated from differences in cohort composition.

4. Statistical analysis

Counts per million used the library sum over the full expression matrix, followed by $\log_2(\mathrm{CPM}+1)$. This transformation is sample-local and pools no information across patients. A transcript was measurable if CPM reached 1 in at least 20% of training tumours. Each training partition separately determined median imputation, removal of constant features, and mean/standard-deviation scaling. Missing annotation was never replaced with zero. No batch correction was applied across folds.

The panel used an elastic-net penalised Cox model over a penaliser grid of {0.001, 0.01, 0.1, 1, 10} and an L1-ratio grid of {0.1, 0.5, 0.9, 1.0}. Nested cross-validation used five outer and three inner folds stratified by event status, repeated with seeds 11, 23, 37, 53 and 71. Inner-fold comparable-pair-weighted concordance selected hyperparameters, with ties favouring stronger penalisation and then a larger L1 ratio. Standardised coefficients below 10^{-4} in absolute value were treated as zero and the baseline hazard recomputed. A separate full-cohort refit summarised retained coefficients and was never used to score held-out discrimination.

Harrell's C-index compared risks only within each held-out outer fold. Comparable-pair counts were aggregated across folds within a seed, and the five seed-specific estimates averaged. Cumulative/dynamic AUC used inverse-probability-of-censoring weighting with the censoring distribution estimated on the corresponding training set. A horizon was reported only if, in every outer fold, there were at least three held-out events by the horizon, five held-out observations beyond it, ten training observations beyond it, and training censoring survival of at least 0.05. A horizon failing these rules was reported as

not estimable rather than averaged over the folds that happened to pass — a conservative reporting rule, not a claim that every alternative estimator is undefined.

For the random comparator, 1,000 panels of four distinct genes were drawn uniformly from the full annotation universe after excluding the four candidates, and each was put through the identical nested cross-validation, measurement gate, preprocessing and hyperparameter search. The same draws were reused across cohorts. Genes were not prefiltered on whole-cohort expression nor matched to the candidates by expression level. Percentiles used midranks; the one-sided empirical probability was (1 + #{C_random ≥ C_panel})/1001.

Each transcript was also fitted alone in an unpenalised continuous-expression Cox model on exactly the panel's outer partitions, with gate, imputation and scaling fitted on the corresponding training set. Reusing the folds permits paired comparison but does not correct the optimism of choosing the strongest marker after inspecting results.

4.1 The random single-gene null, added after submission

The null of §10 was not part of the submitted study. It draws 1,000 genes uniformly from the 60,656 remaining after the four candidates are removed, and puts each through the author's own frozen fold assignments, measurement gate, training-only imputation and scaling, unpenalised univariate Cox fit and concordance rule. A coefficient below 10^{-4} in absolute value is set to zero, which is the author's own retention convention and the reason a model that fits nothing scores exactly 0.500 rather than an arbitrary value.

Neither `lifelines` nor `scikit-survival` could be installed in the environment available, so the per-fold Cox fit and the concordance counts were re-implemented, with Efron handling of tied event times. The re-implementation was then required to reproduce the author's recorded results before it was allowed to produce a null: it returns all four published single-marker C-indices to the last decimal place recorded, together with all 100 per-fold C-indices, comparable-pair counts and coefficients across the four markers and 25 folds, and it reproduces the archived per-sample library sizes exactly. The expression matrix was re-downloaded and its SHA-256 checked against the hash recorded in the author's `results.json`, so the input is the analysed snapshot and not a later revision.

The matched null restricts the 1,000 draws to those passing the measurement gate in all 25 outer folds, which all four candidate transcripts do. That criterion uses expression only and no outcome, so it introduces no selection on the quantity being tested. Because LINC00675 was reported as the best of four candidates, the selection-aware probability is the chance that the maximum of four genes drawn from the matched null reaches its C-index, estimated from 200,000 resamples.

Percentile 95% confidence intervals used 1,000 patient-cluster bootstrap replicates, each patient's multiplicity shared across all repeated held-out predictions. These intervals condition on the fitted models and training censoring distributions: model fitting, hyperparameter tuning and marker selection were **not** repeated inside the bootstrap, so the intervals understate total uncertainty.

5. The panel did not beat its own null

The primary panel reached a held-out C-index of 0.609 (95% CI, 0.509–0.717), below the 0.65 target fixed in advance. It ranked at the 93.4th percentile of the 1,000 random four-gene panels, one-sided empirical $p = 0.067$. The mean random-panel C-index was 0.502. All 1,000 random panels were retained in the comparison; two failed inner candidate fits in a single random panel were handled by invalidating that hyperparameter candidate rather than dropping the panel.

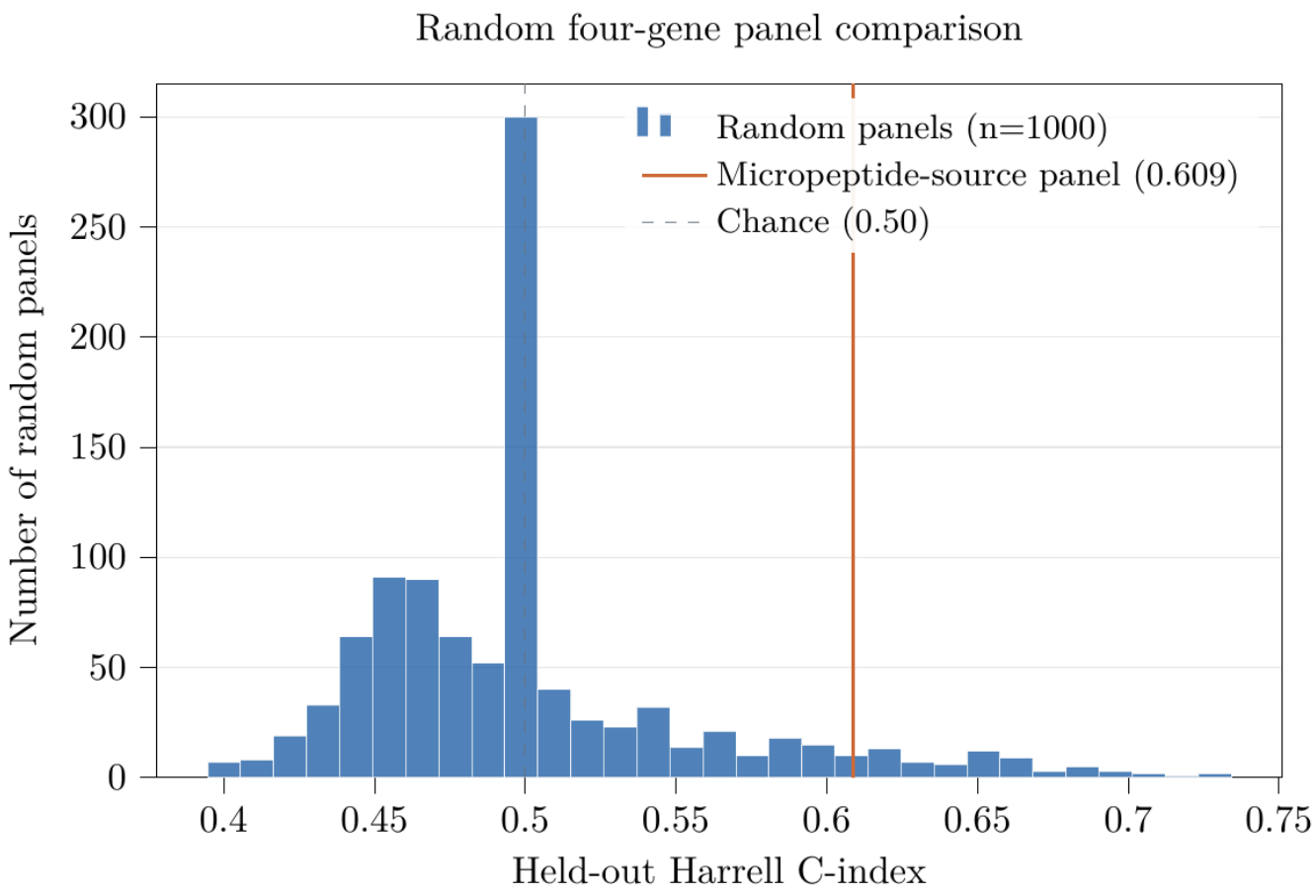


Fig. 1 | Primary-DFI C-indices for 1,000 random four-gene panels put through the identical nested cross-validation. The line marks the micropeptide-source panel at 0.609, the 93.4th percentile, one-sided empirical p = 0.067. The spike at exactly 0.500 is 264 panels whose coefficients were all shrunk to zero, leaving constant risk; §5.1 reports the comparison restricted to the 736 panels that fitted something.

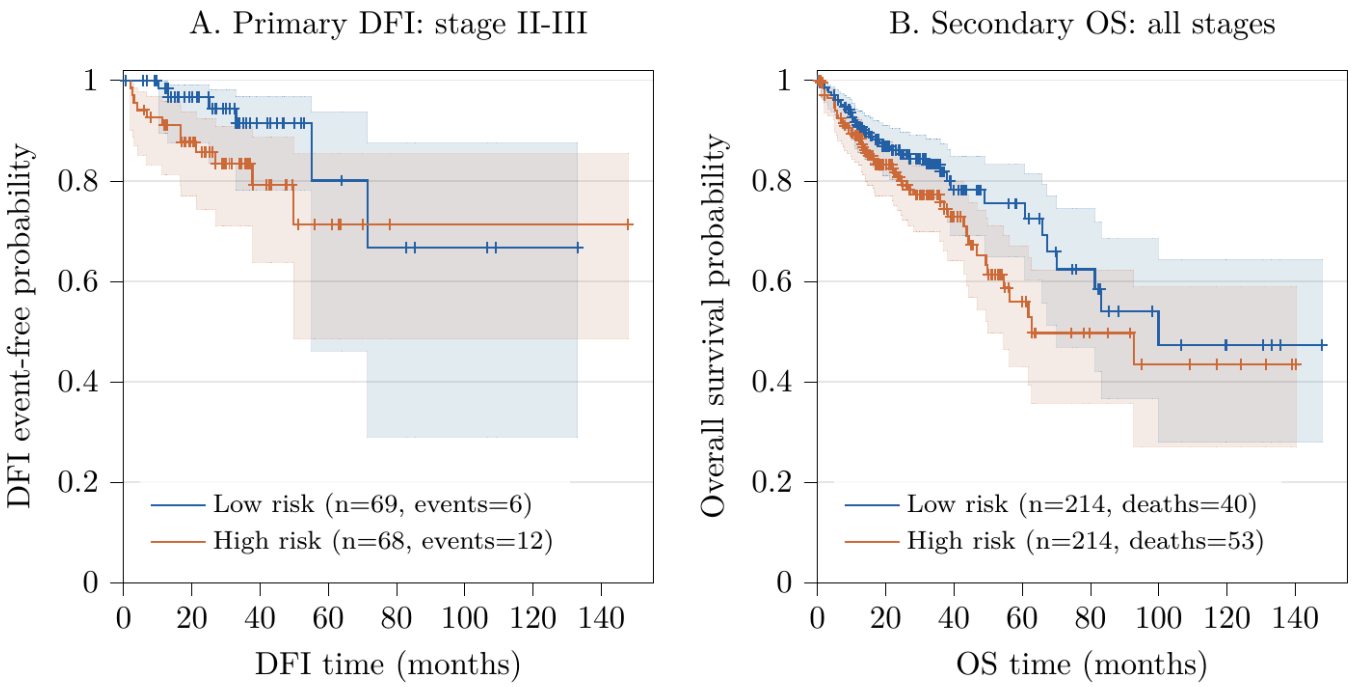


Fig. 2 | Descriptive Kaplan–Meier curves by four-transcript panel risk for (a) primary DFI, stages II/III, and (b) secondary overall survival, stages I–IV. Each held-out patient's risk percentile was computed against the corresponding outer-training distribution, averaged across repetitions and split at the cohort median. Shading is pointwise 95%; ticks mark censoring. Not a validated risk stratification.

The same conclusion holds in both secondary cohorts, with larger probabilities: 0.134 for expanded DFI and 0.102 for all-stage OS. A mechanistically chosen panel that cannot separate itself from randomly drawn genes put through the same machinery has not demonstrated that the mechanism carries prognostic information in this dataset.

5.1 A quarter of the null distribution is inert

The null in Fig. 1 has a conspicuous spike at exactly 0.500. It is not a rendering artefact. Of the 1,000 random panels, **264 (26.4%) returned a C-index of exactly 0.500**, because for those panels the selected penalty shrank every coefficient to zero, leaving a constant risk score and therefore, by definition, a concordance of one half. These panels are not weak competitors; they are models that declined to fit.

This matters for the headline comparison. The published probability treats all 1,000 draws as comparators, giving 66 at or above the panel and $p = 0.067$. Restricted to the 736 panels that fitted anything at all, 66 of 736 (9.0%) match or beat the candidate panel, a one-sided probability of 0.091. The candidate panel's rank is flattered by a comparator distribution that is a quarter inert.

Both figures exceed 0.05 and the prespecified conclusion is unchanged, so this does not rescue or reverse anything. It is reported because the percentile is the paper's headline comparison and a reader is entitled to know that roughly a quarter of the distribution it is measured against consists of null models in the strict sense. The effect works against the panel, which is why the negative result should be read as firmer rather than more fragile.

6. What the elastic net kept

Only LINC00675/TMEM238L survived the final full-cohort refit, at penaliser 0.1 and L1 ratio 0.5. That final model must be distinguished from the models that produced the held-out predictions: across the 25 outer folds, LINC00675 was retained in 23, HOXB-AS3 in 18, LINC00467 in 16 and ELFN1-AS1 in 12. All four passed the measurement gate in every fold. The panel's C-index therefore describes a collection of fold-specific penalised models, not repeated fitting of one fixed model.

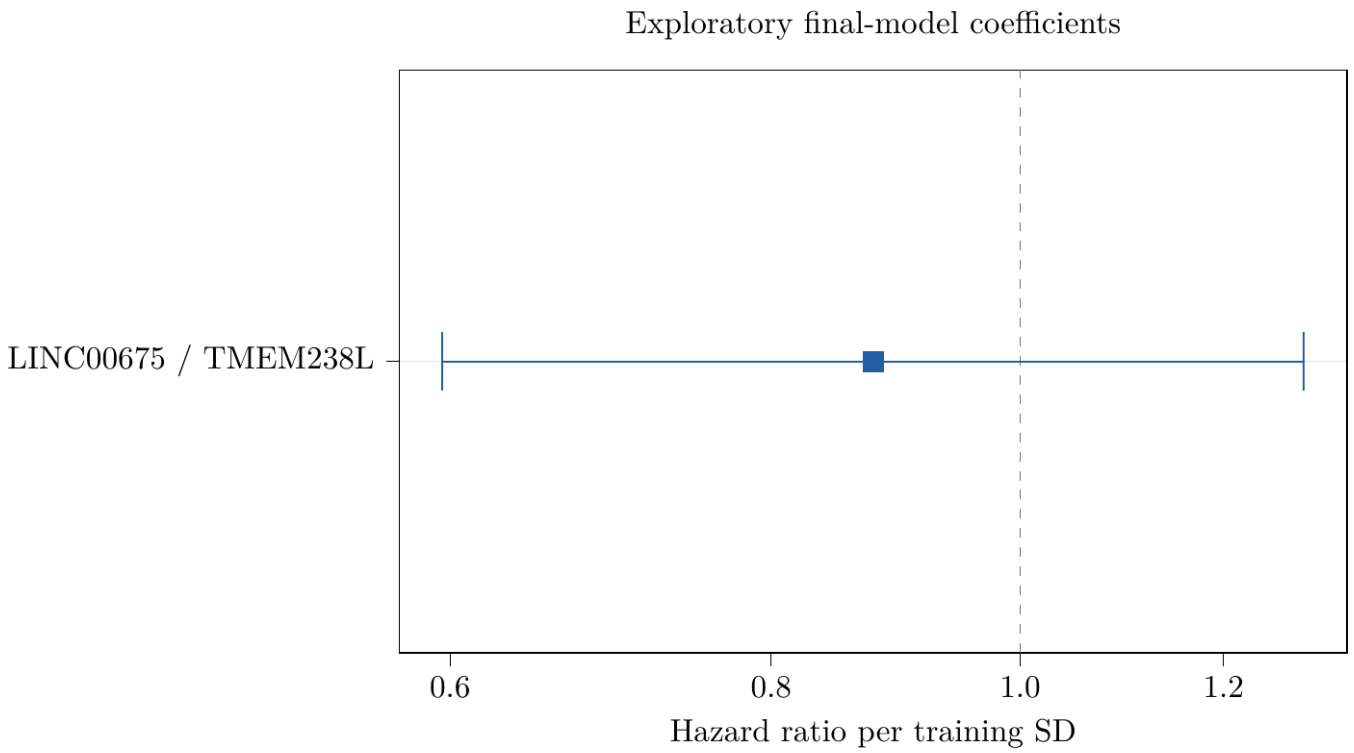


Fig. 3 | Coefficient from the final full-cohort penalised refit, in which LINC00675/TMEM238L was the only transcript retained. The hazard ratio is per standardised expression unit, and the Wald interval ignores both variable selection and penalisation, so it is not confirmatory. This refit produced none of the held-out estimates in this paper.

7. Three of the four transcripts were at or below chance

The single-marker benchmarks are the most informative part of this study, and they are not encouraging for the panel concept.

Source RNA	Primary DFI (II/III)	All-stage DFI	All-stage OS
LINC00467	0.462 (0.343–0.582)	0.400 (0.314–0.488)	0.431 (0.394–0.469)
ELFN1-AS1	0.367 (0.305–0.436)	0.479 (0.353–0.596)	0.588 (0.519–0.655)
LINC00675 / TMEM238L	**0.700 (0.583–0.819)**	0.691 (0.571–0.802)	0.502 (0.435–0.563)
HOXB-AS3	0.515 (0.380–0.652)	0.573 (0.423–0.715)	0.451 (0.420–0.484)

For primary DFI, three of four transcripts sat at or below chance. Across all twelve transcript-by-cohort combinations, seven estimates fall below 0.5, and LINC00467 is below 0.5 in all three cohorts. Choosing transcripts because their peptides have documented colorectal biology did not, in this dataset, yield transcripts that carry prognostic signal.

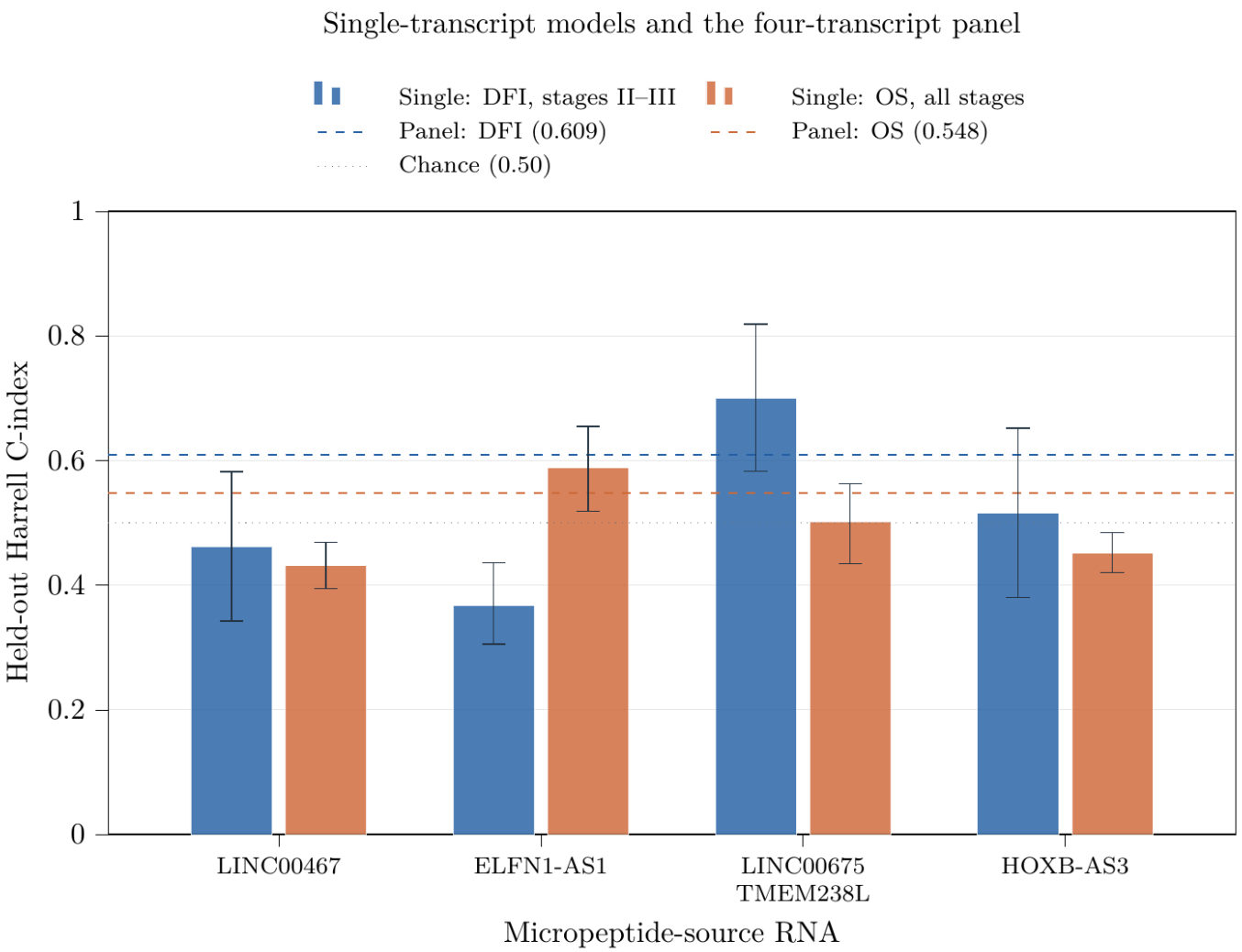


Fig. 4 | Held-out single-transcript Cox discrimination for primary DFI and all-stage overall survival, with 95% intervals from 1,000 patient-cluster bootstrap replicates of held-out predictions without refitting. Dashed lines mark the corresponding panel estimates. Three of four transcripts sit at or below chance for DFI, and the DFI and OS orderings invert.

Against that background, LINC00675's 0.700 for primary DFI is the single exception among four mechanistically motivated candidates. The paired panel-minus-LINC00675 difference was −0.092 (95% CI, −0.192 to −0.007), conditional on the fitted models: the four-transcript strategy was worse than its own best component. That contrast is post hoc, uncorrected for multiplicity, and conditional on models the bootstrap did not refit.

8. Stage, endpoint and horizon support

In the 181-patient expanded DFI cohort with 21 events, LINC00675 reached 0.691 (95% CI, 0.571–0.802) against a panel C-index of 0.593. This cohort contains the primary cohort and is a stage-sensitivity analysis, not independent replication.

The endpoint contrast is the sharpest internal check available, and LINC00675 does not survive it. In the 428-patient OS cohort with 93 deaths — more than five times the primary cohort's events — LINC00675 achieved 0.502 (95% CI, 0.435–0.563), indistinguishable from chance. ELFN1-AS1, which was the *worst* DFI marker at 0.367, became the best OS marker at 0.588. A marker ordering that inverts between endpoints in the same resource is what one expects from noise, though cohort composition, mortality and treatment also differ between the two analyses and cannot be separated here.

Cohort	Horizon (yr)	Events	Beyond horizon	Censored by horizon	AUC (95% CI)
Primary DFI	3	14	46	77	Not estimable
Primary DFI	5	17	13	107	Not estimable
All-stage DFI	3	17	55	109	Not estimable
All-stage DFI	5	20	16	145	Not estimable
All-stage OS	3	70	109	249	0.543 (0.459–0.620)
All-stage OS	5	83	45	300	Not estimable

Only three-year OS AUC met the support rules, at 0.543 (95% CI, 0.459–0.620) — an interval comfortably containing 0.5. Both DFI horizons failed support in 8 and 25 of 25 folds at three and five years; five-year OS failed in one fold, where only four held-out patients remained beyond the horizon. Having more events did not guarantee adequate support in every fold.

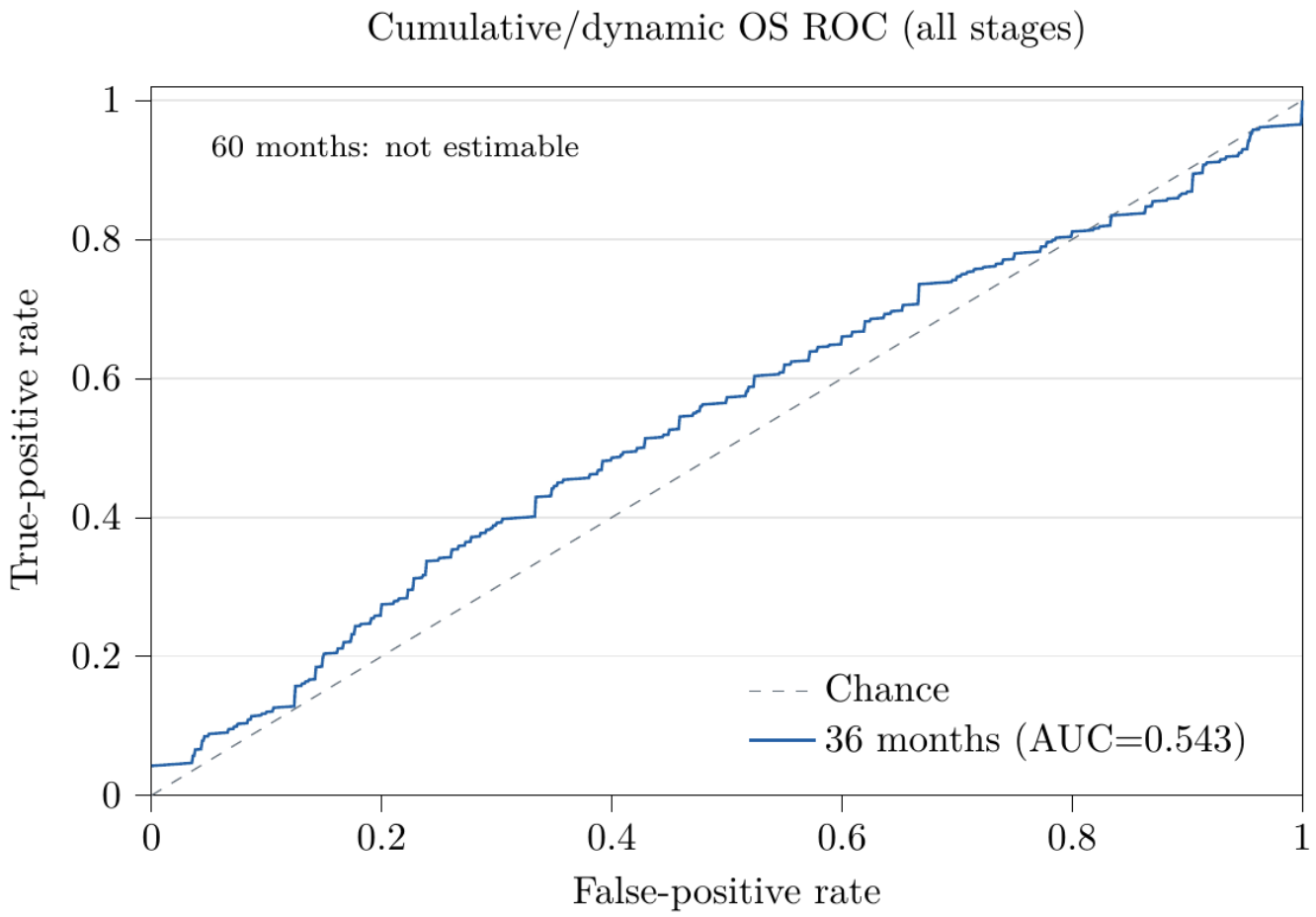


Fig. 5 | Cumulative/dynamic ROC for the panel in all-stage overall survival at three years, the only horizon meeting the fold-support rules, with AUC 0.543 (95% CI, 0.459–0.620). Fold curves were averaged using case–control pair weights within seeds, then across seeds. An internal cross-validation result, not a validation.

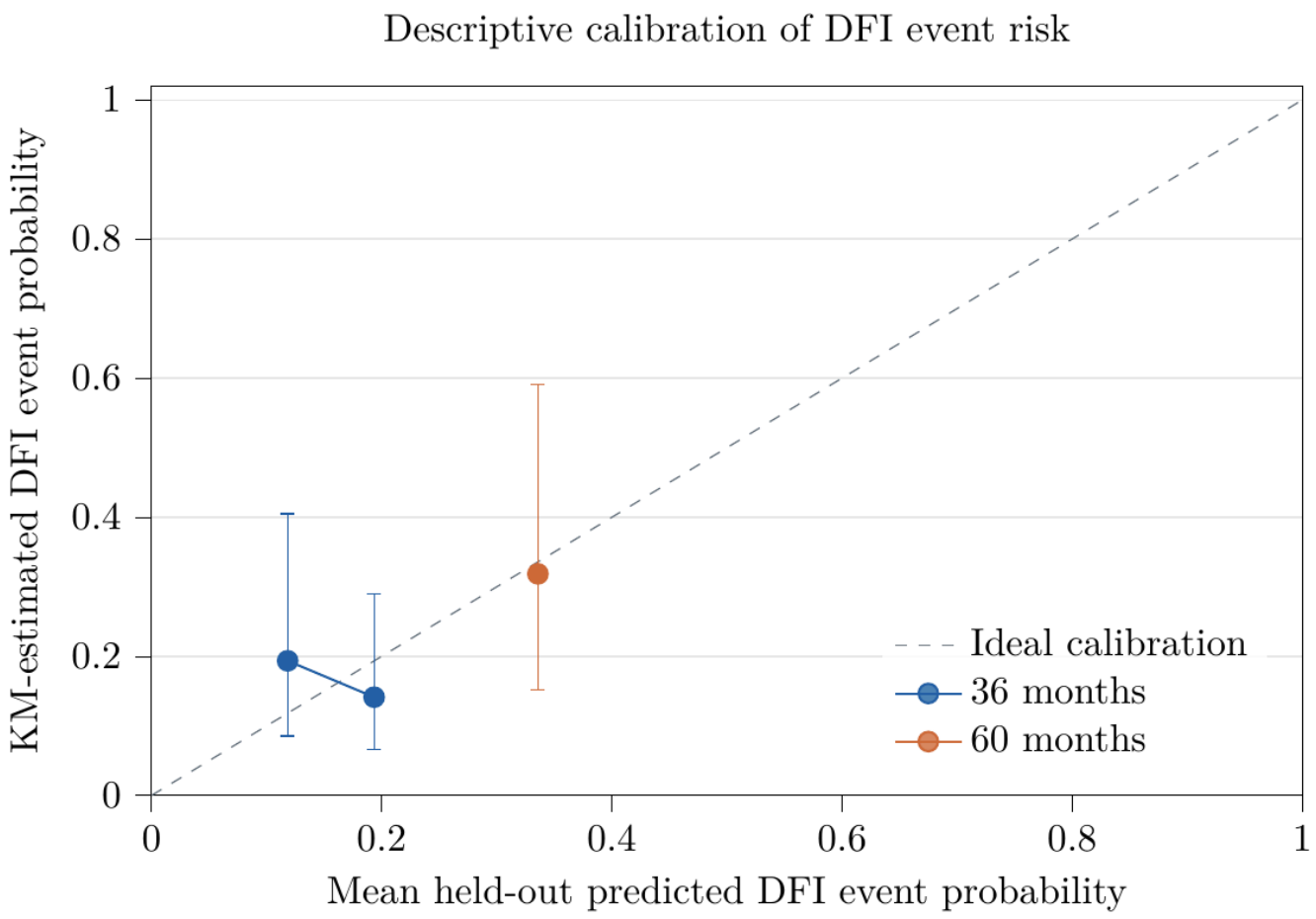


Fig. 6 | Descriptive primary-DFI calibration of the panel. Predicted event probabilities were averaged across held-out repetitions and binned; observed probabilities are Kaplan–Meier estimates with pointwise 95% intervals, and unsupported bins are omitted. These bin criteria differ from the AUC support rules, and the display does not establish reliable clinical calibration.

9. The median split

The primary-cohort expression median was 1.14059 on the $\log_2$(CPM+1) scale. The low-expression group held 69 patients and 12 DFI events; the high-expression group held 68 patients and 6 events. The unadjusted log-rank comparison was not significant (p = 0.115), and the curves cross during sparse late follow-up.

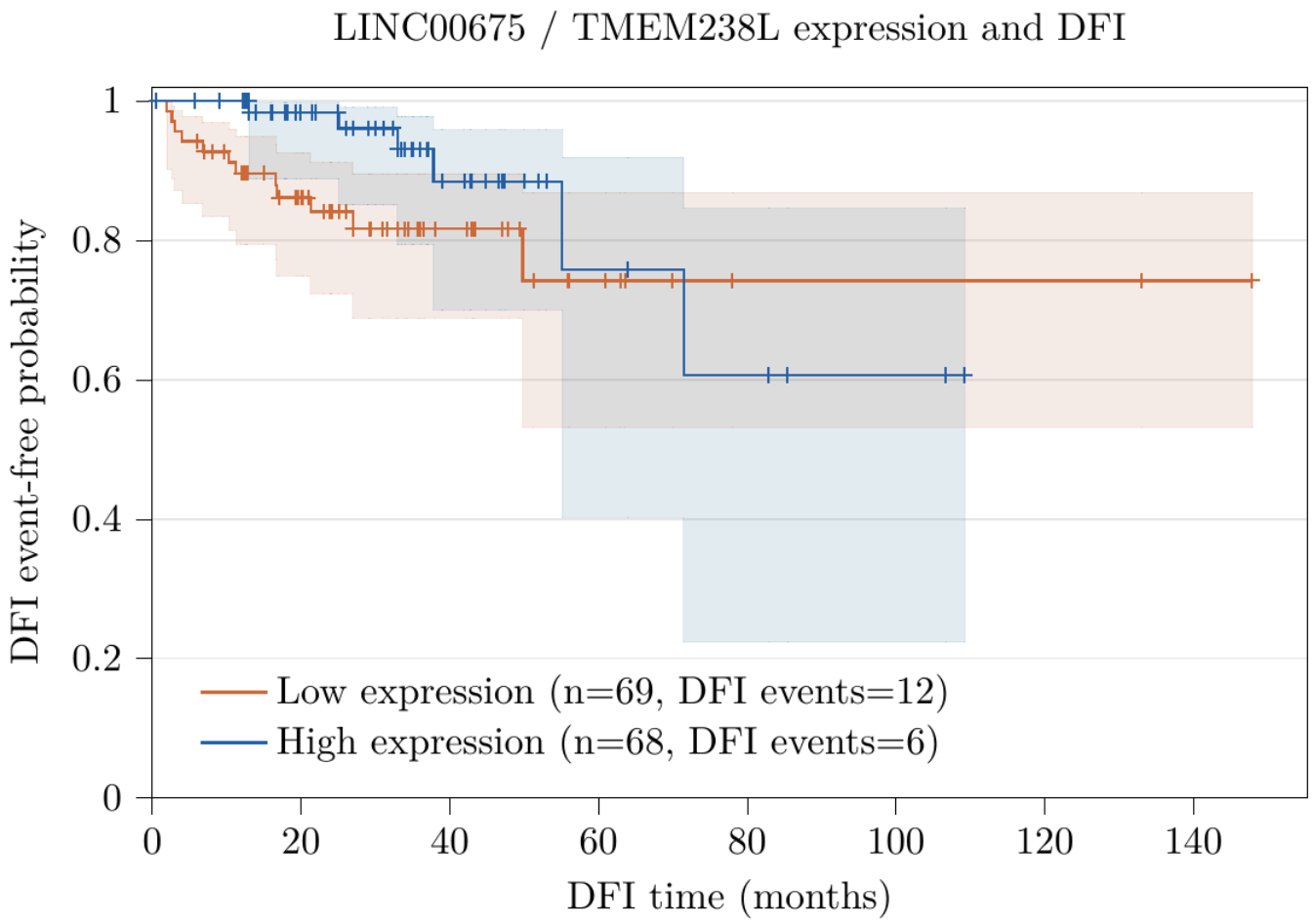


Fig. 7 | Post hoc LINC00675 median split in the primary cohort: 69 patients and 12 events at or below the median, 68 patients and 6 events above it; log-rank p = 0.115. The cutoff was computed on the whole cohort for display only and never entered Cox fitting or cross-validated scoring. The curves cross during sparse late follow-up.

This cutoff was computed on the whole cohort for display only and never entered Cox fitting or cross-validated scoring. A dichotomised whole-cohort comparison and continuous-expression held-out discrimination answer different questions, and failing the first does not invalidate the second. But a reader looking for confirmation that LINC00675 separates recurrence will not find it here either.

10. The comparison the study was missing

The decisive comparison was absent from the submitted manuscript, and it is the one this version adds.

The panel had a null: 1,000 random four-gene panels through the same pipeline. LINC00675 had none. There was therefore no calibrated answer to the only question that

matters about 0.700 — how often does a single gene drawn at random, fitted the same way on the same 137 patients and the same 25 folds, discriminate this well? The methods of §4.1 describe how that null was built and the checks it had to pass first.

Against all 1,000 random genes, LINC00675 sits at the 99.7th percentile: 3 draws of 1,000 reach 0.700, a one-sided empirical probability of 0.004. That figure overstates the case, and the reason is the same one that distorted the panel comparison. Of the 1,000 single genes, **710 produce no model at all** — they fail the measurement gate, or their fitted coefficient is numerically zero — and score exactly 0.500 by construction. A null that is 71% inert is not a fair opponent.

Restricting to genes that could have been candidates fixes this. Of the 1,000 draws, 279 pass the measurement gate in all 25 outer folds, which is what all four candidate transcripts do. This subset is selected on expression alone, with no reference to outcome. Its mean C-index is 0.510 and its median 0.503, and none of it is degenerate.

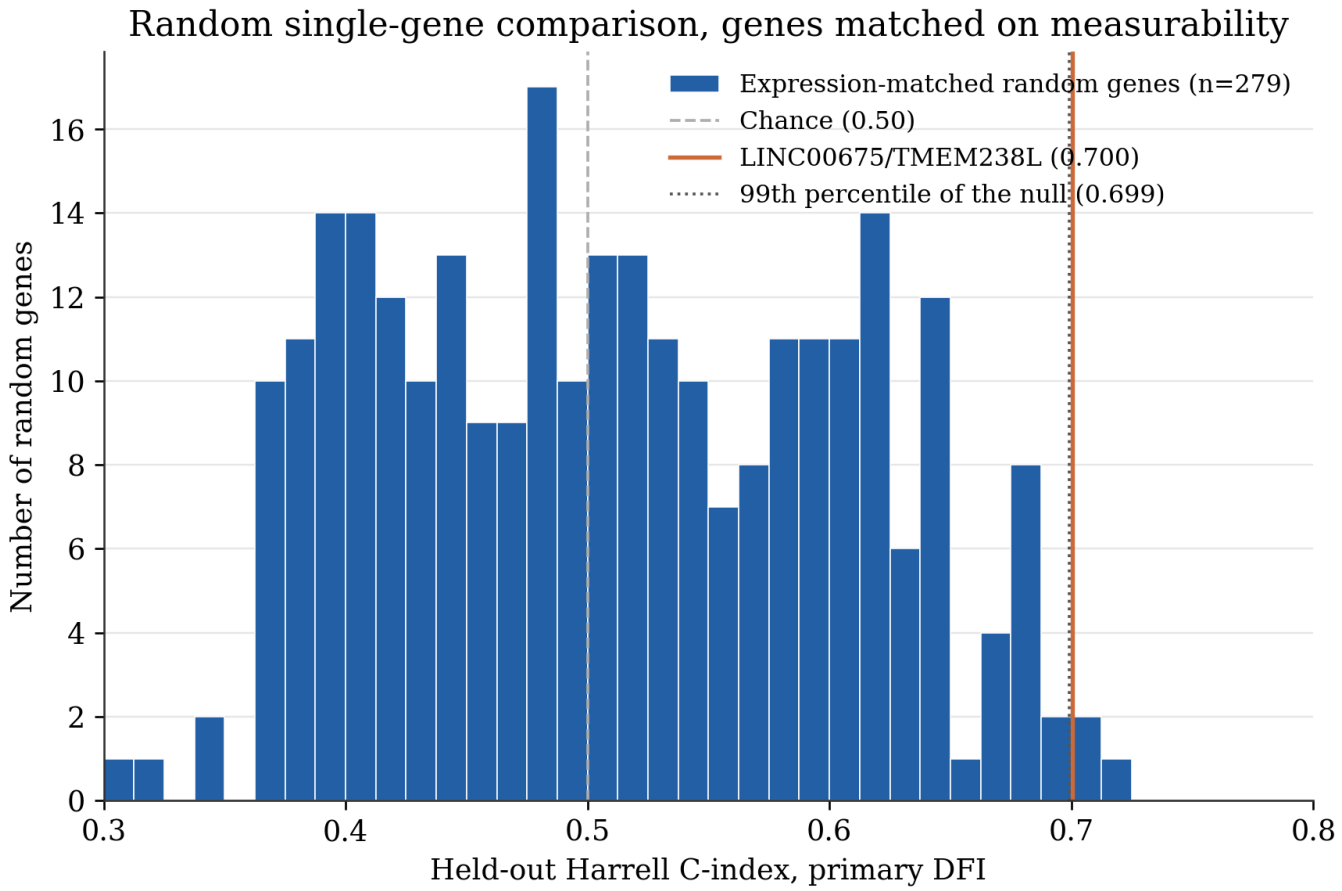


Fig. 8 | The random single-gene null the submitted study lacked, added by the journal. Of 1,000 genes drawn at random and put through the author's own folds, gate, model and scoring, the 279 that pass the measurement gate in all 25 folds — as all four candidate transcripts do — form the comparator shown. LINC00675/TMEM238L sits at the 98.9th percentile, just above the null's 99th percentile of 0.699. Correcting for its having been the best of four candidates, a result this good arises by chance 4.2% of the time. The re-implementation used to build this null first reproduced all four of the author's published single-marker C-indices and all 100 of his per-fold values exactly.

Within that matched null, 3 of 279 genes reach 0.700, a one-sided probability of **0.014**, placing LINC00675 at the 98.9th percentile. The null's own 99th percentile is 0.6993; LINC00675 scores 0.7005. It is, by a small margin, the kind of result you see about once in a hundred random genes.

Correcting for selection is the last step and the one that matters most, because LINC00675 was not drawn at random — it was the best of four candidates. The relevant probability is therefore that the maximum of four genes drawn from the matched null reaches 0.700, which across 200,000 resamples happens **4.2%** of the time.

So the answer to the question this study could not previously ask is: yes, but barely. A selection-aware probability of 0.042 clears the conventional threshold with almost

nothing to spare, on 18 events, in the cohort where the marker was found. It is enough to justify the cost of an external test. It is not enough to call LINC00675 a recurrence biomarker, and it says nothing about whether the marker adds information to stage, mismatch-repair status, age and treatment, which remains the question that decides clinical usefulness.

11. What this study establishes

A panel of four micropeptide-source RNAs, chosen on documented peptide biology rather than by screening, does not predict disease-free interval in TCGA colon adenocarcinoma well enough to pursue. It missed a target set in advance and did not separate from randomly drawn panels under an identical procedure, in the primary cohort and in both secondary ones. That is a clean prespecified negative result, and it is the finding of this paper.

Three further things follow, none of them a biomarker claim. First, mechanistic selection did not concentrate prognostic signal: most of these transcripts discriminated at or below chance. Second, combining four transcripts was worse than using the best one, which identifies a failed strategy without identifying its statistical cause — the final refit kept only LINC00675, but the held-out predictions came from fold-specific models that often retained others, and feature inclusion, fitted direction and risk ties are all plausible contributors. Third, LINC00675 now has a calibrated standing it did not have before: against genes that could have been candidates, and after correcting for having been the best of four, a C-index of 0.700 arises by chance about 4% of the time.

That is the strongest statement the data support, and it is deliberately a weak one. A selection-aware probability of 0.042 on 18 events is a reason to run one external test, not a result. What this study still does not establish is that LINC00675 predicts recurrence: it shows no signal for overall survival, does not separate recurrence on a median split, has never been seen outside this cohort, and has never been tested against stage, mismatch-repair status, age or treatment. Bulk abundance of this transcript could track epithelial differentiation or tumour cell fraction with no change in peptide abundance at all. RNA is not peptide: no immunohistochemistry or targeted mass spectrometry was performed, and nothing here can establish a FORCP-mediated mechanism.

12. Limitations

Eighteen events in the primary cohort is the limitation from which most others follow: unstable fold estimates, failed horizon support, and wide intervals. Excluding 168 of 305 eligible patients for missing DFI status may also introduce selection bias of unknown direction.

The bootstrap intervals condition on already-fitted cross-validation models and do not capture variability from model fitting, hyperparameter selection or post hoc marker choice. The repeated folds overlap and are not independent replications. Proportional-hazards adequacy was not assessed.

The single-gene null has limits of its own. It reuses the same 25 folds and the same 137 patients, so it calibrates against sampling variation in this cohort rather than against a new one, and it cannot substitute for external data. Matching on the measurement gate equalises measurability but not expression level, dynamic range or correlation structure, so the matched comparator is fairer than the unmatched one without being an exact like-for-like. The best-of-four correction treats the four candidates as four independent draws, which is conservative only if they are uncorrelated; any positive correlation between them would make the correction slightly too harsh, and any further undeclared looking would make it too lenient. Stage, age, treatment and mismatch-repair status were not added as covariates, so incremental value beyond clinical risk was never tested — and for a prognostic marker that is the question that decides clinical usefulness.

No external validation was performed. The expanded DFI cohort overlaps the primary cohort and the OS analysis draws on the same resource, so neither is independent. The random null consists of four-gene panels drawn without expression matching; it contextualises the panel and provides neither a single-marker test nor a substitute for validation.

Gene-level measurement cannot distinguish coding isoforms, translation efficiency or protein turnover, and the historical lncRNA label should not be read as a contemporary coding classification for TMEM238L. DFI is used exactly as supplied by the Clinical Data Resource and is not interchangeable with similarly named endpoints [12].

TCGA's racial imbalance is documented [13]. In these cohorts White patients were a majority among those with recorded race, and race was unavailable for many. Generalisability across populations and care settings is untested, and the study is confined to colon adenocarcinoma; nothing here extends to rectal cancer.

13. What would settle it

The cheap screening step is done. LINC00675 passed it narrowly, which is the outcome that justifies spending more and does not justify believing the result.

A locked external test of LINC00675 alone is next. GSE39582 is a practical candidate only if its array probes measure TMEM238L adequately and uniquely; probe mapping, preprocessing, outcome definition and the continuous-marker model must all be fixed in writing before any outcome is examined. Given a selection-aware probability of 0.042 in the discovery cohort, a failed external test should be treated as the more likely outcome and reported either way.

The decisive result is not the external association on its own. It is whether LINC00675 adds reproducible information beyond stage, mismatch-repair status, age and treatment. Reproducing an unadjusted association would leave the clinical question exactly where it is now.

One cheaper check is also worth doing first, because it could remove the marker without any new cohort: whether bulk LINC00675 abundance is explained by tumour purity or by an epithelial-differentiation score computed on the same specimens. If it is, the association has an explanation that has nothing to do with FORCP.

Declarations

Author. Leo Lee is the sole author. He designed the study, specified the candidate panel and endpoints, ran and audited the analysis pipeline, and approves this version.

Funding. None.

Competing interests. None declared by the author.

Ethics. Not applicable. The study analyses de-identified public data from TCGA and the TCGA Clinical Data Resource. No human participants were recruited and no new specimens were collected.

Use of artificial intelligence. This paper was prepared with substantial AI assistance at every stage, and the division of labour is set out here rather than summarised.

For the underlying study, as disclosed by the author: OpenAI ChatGPT (GPT-5) assisted with literature retrieval and with identifying candidate micropeptide-source RNAs; OpenAI Codex assisted with writing and debugging the Python pipeline, including CPM normalisation, nested cross-validated elastic-net Cox regression, the random-panel baseline, bootstrap intervals and figure export. The author states that all code was executed and audited by him and that all cited references were verified against the original publications. The archived package records these as author-supplied attestations that its own automated audit did not independently verify, and they are published here on that basis.

For this version: the manuscript submitted to PRISM carried an AI-assisted Abstract, Discussion and Conclusion together with a declaration that the author would independently rewrite those sections before submission. That rewrite had not been performed — the archived validation record states `human_rewrite_completed: false` — and it has still not been performed. Instead, Claude (Anthropic), acting for the journal, reframed the paper for publication: the title, abstract, section structure, the emphasis in §§5–13 and all prose in those sections are AI-written. No AI system is an author.

None of the author's results were altered. No model of his was refitted and no reported number of his was changed; the Methods describe the same pipeline as the submitted version.

Four quantitative additions are new to this version and were made by the journal. Three are arithmetic on the author's frozen outputs: the count of degenerate random panels and the restricted probability in §5.1, the tally of below-chance estimates in §7, and the independent reproduction of his published empirical p of 0.066933 from `random_panel_scores.tsv` as a check that his frozen outputs and his reported numbers agree.

The fourth is new computation: the random single-gene null of §10 and §4.1, with Fig. 8. This required re-downloading the expression matrix, whose SHA-256 was checked

against the hash in the author's `results.json`, and re-implementing the per-fold Cox fit and concordance count because `lifelines` and `scikit-survival` could not be installed. That re-implementation was gated on reproducing the author's recorded results: all four published single-marker C-indices to the last recorded decimal, all 100 per-fold C-indices, pair counts and coefficients, and the archived per-sample library sizes. It is the journal's computation, not the author's, and the code is in the deposit so that it can be checked or discarded independently of his.

Peer review. This paper is published as a working paper and has not been peer reviewed. See the note on the article page.

Data and code availability

The study uses the public TCGA-COAD expression matrix from UCSC Xena and the TCGA Clinical Data Resource. Neither is redistributed here. The journal's addition — the single-gene null code, the 1,000 sampled C-indices, the gate mask and the figure script — is in `prism-single-gene-null/` inside the deposit, alongside its validation output. It is separate from the author's material and can be checked or rejected on its own.

The author's reproducibility package is deposited with the journal as coad-micropeptide-package.zip (30 MB). It contains the analysis pipeline, the frozen results, the patient-filtering audits, the cross-validation splits, the random-panel outputs and the figure sources, with SHA-256 hashes for the two required inputs. It is journal-hosted rather than archived, so it carries no DOI and no persistent identifier. The two source matrices are excluded and must be downloaded from the addresses above; the package records their hashes so a rerun can be checked against the analysed snapshot. Proposed validation data are at GEO accession GSE39582.

Figures are renumbered in this version to follow the argument. The correspondence to the archived package is: Fig. 1 here is fig4_null; Fig. 2 is fig1_km; Fig. 3 is fig3_forest; Fig. 4 is fig6_single; Fig. 5 is fig2_roc; Fig. 6 is fig5_calib; Fig. 7 is fig7_single_km. Fig. 8 has no counterpart in the archive: it is the journal's addition and is generated by the deposited null code. The figures themselves are unaltered, recompiled from the author's TikZ/PGFPlots sources.

References

1. Chothani, Sonia P.; Adami, Eleonora; Widjaja, Anissa A.; et al. (2022). A high-resolution map of human RNA translation. *Molecular Cell* 82(15), 2885–2899.e8. DOI: 10.1016/j.molcel.2022.06.023.

2. Shi, Chengyu; Liu, Fangzhou; Su, Xinwan; et al. (2025). Comprehensive discovery and functional characterization of the noncanonical proteome. *Cell Research* 35(3), 186–204. DOI: 10.1038/s41422-024-01059-3.

3. Deutsch, Eric W.; Kok, Leron W.; Mudge, Jonathan M.; et al. (2026). Expanding the human proteome with microproteins and peptideins. *Nature* 654, 813–825. DOI: 10.1038/s41586-026-10459-x.

4. Ge, Qiwei; Jia, Dingjiacheng; Cen, Dong; et al. (2021). Micropeptide ASAP encoded by LINC00467 promotes colorectal cancer progression by directly modulating ATP synthase activity. *Journal of Clinical Investigation* 131(22), e152911. DOI: 10.1172/JCI152911.

5. Zheng, Caishang; Wei, Yanjun; Zhang, Peng; et al. (2023). CRISPR–Cas9-based functional interrogation of unconventional translatome reveals human cancer dependency on cryptic non-canonical open reading frames. *Nature Structural & Molecular Biology* 30(12), 1878–1892. DOI: 10.1038/s41594-023-01117-1.

6. Li, Xiao Ling; Pongor, Lőrinc; Tang, Wei; et al. (2020). A small protein encoded by a putative lncRNA regulates apoptosis and tumorigenicity in human colorectal cancer cells. *eLife* 9, e53734. DOI: 10.7554/eLife.53734.

7. Huang, Jin-Zhou; Chen, Min; Chen, De; et al. (2017). A peptide encoded by a putative lncRNA HOXB-AS3 suppresses colon cancer growth. *Molecular Cell* 68(1), 171–184.e6. DOI: 10.1016/j.molcel.2017.09.015.

8. Zhu, Song; Wang, Ji-Zhong; Chen, De; et al. (2020). An oncopeptide regulates m⁶A recognition by the m⁶A reader IGF2BP1 and tumorigenesis. *Nature Communications* 11, 1685. DOI: 10.1038/s41467-020-15403-9.

9. Marisa, Laetitia; de Reyniès, Aurélien; Duval, Alex; et al. (2013). Gene expression classification of colon cancer into molecular subtypes: characterization, validation, and prognostic value. *PLoS Medicine* 10(5), e1001453. DOI: 10.1371/journal.pmed.1001453.

10. Gray, Richard G.; Quirke, Philip; Handley, Kelly; et al. (2011). Validation study of a quantitative multigene reverse transcriptase-polymerase chain reaction assay for assessment of recurrence risk in patients with stage II colon cancer. *Journal of Clinical Oncology* 29(35), 4611–4619. DOI: 10.1200/JCO.2010.32.8732.

11. Goldman, Mary J.; Craft, Brian; Hastie, Mim; et al. (2020). Visualizing and interpreting cancer genomics data via the Xena platform. *Nature Biotechnology* 38(6), 675–678. DOI: 10.1038/s41587-020-0546-8.

12. Liu, Jianfang; Lichtenberg, Tara; Hoadley, Katherine A.; et al. (2018). An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell* 173(2), 400–416.e11. DOI: 10.1016/j.cell.2018.02.052.

13. Spratt, Daniel E.; Chan, Tiffany; Waldron, Levi; et al. (2016). Racial/ethnic disparities in genomic sequencing. *JAMA Oncology* 2(8), 1070–1074. DOI: 10.1001/jamaoncol.2016.1854.